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Development and application of data-driven CHF model in system analysis code

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ABSTRACT

In nuclear reactor systems, when the fuel rods reach the critical heat flux (CHF), a sharp increase in fuel temperature occurs due to a drastic reduction in heat transfer capacity, thus posing a considerable risk to the reactor's safety. Consequently, accurate CHF prediction holds paramount importance in accurately simulating accident scenarios and enhancing the overall safety of reactor systems. To tackle the challenge of limited prediction accuracy in existing CHF models, this study initially established a comprehensive database by utilizing available experimental data and lookup tables. Subsequently, various methodologies, including the Back Propagation Neural Network (BPNN), Random Forest (RF), and Physics-Informed Machine Learning (PIML), were employed to develop multiple CHF prediction models, and their performance was thoroughly evaluated. Furthermore, the optimal CHF model was integrated into the self-developed analysis code ARSAC, which was then validated using the ORNL-THTF experiment. The results indicated that the BPNN-based model not only demonstrated exceptional prediction accuracy but also exhibited rapid calculation speeds. Notably, the average relative error between the experimental data points and the calculation results of the modified code is 3.64%, while for the original code, it is 22.84%. This study effectively leverages the strengths of data-driven approaches, providing a robust technical solution for high-precision, efficient, and adaptive numerical prediction and analysis of reactor accidents.

1. Introduction

Under normal operating conditions of pressurized water reactor (PWR), the heat generated in the reactor core is dissipated by the single-phase coolant. Nevertheless, in the event of accidents like loss-of-flow, the coolant within the core undergoes a phase transition, forming the two-phase gas–liquid mixture. Under these circumstances, the core may experience heat transfer degradation, leading to a sharp increase in fuel temperature, which poses a significant threat to reactor safety, including the risk of core melting. Critical heat flux (CHF) denotes the threshold at which the boiling mechanism begins transitioning from nucleate boiling to film boiling on the surface of fuel elements. CHF serves as a pivotal parameter in identifying the onset of heat transfer degradation. Accurate calculation of CHF is imperative for precisely anticipating alterations in critical thermal–hydraulic parameters during accidental scenarios, safeguarding fuel temperatures from exceeding safety thresholds, and bolstering the overall safety of the reactor system. Additionally, it aids in avoiding overly cautious designs that might undermine the economic viability of the reactor system.

The mechanism of CHF is complex and influenced by factors such as heat flux distribution, hydraulic diameter, fluid velocity, pressure, among others. Numerous researchers have conducted experimental and theoretical analyses on CHF. Chen et al. (Chen et al., 2023) investigated the CHF of a vertical annulus channel with a short-heated rod at low pressure and low flow conditions. Zhan et al. (Zhang et al., 2020) investigated the CHF in a vertical tube with an inner diameter of 6.0 mm under inlet flow oscillation conditions. Lyons et al. (Lyons et al., 2020) conducted CHF experiments under high pressure in a 2 × 2 rod bundle channel. Song et al. (Song et al., 2021) rearranged the sublayer dry-out model for DNB and proposed a new mechanism for dry-out and proposed a new CHF prediction correlation with a simple structure but relatively high accuracy. However, experimental studies are typically based on

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specific conditions, limiting the applicability of semi-theoretical, semi-empirical correlations derived from experiments. A single CHF correlation cannot accommodate the wide range of pressure, temperature, and flow parameters in reactor systems. Consequently, in reactor design and analysis, it is necessary to select the appropriate CHF correlation based on the specific scenario. Commonly used CHF correlations (Bruder et al., 2017) and their applicable ranges are shown in Table 1. Given the limited applicability and prediction errors associated with empirical correlations, and the increasing demand for accuracy and applicability in reactor simulations, traditional semi-theoretical, semi-empirical correlations are gradually becoming inadequate for precise CHF predictions.

In recent years, the remarkable advancements in computational and digital technologies have propelled numerical reactor technology to new heights. The numerical reactor is designed to simulate various mechanistic phenomena and predict critical parameters in reactor systems by employing refined modeling techniques and integrating different physical models. Numerous projects, including CASL (The consortium for advanced simulation of light water reactors), NURESIM (NUclear REactors SIMulation), and NEAMS (Nuclear Energy Advanced Modeling and Simulation), have been initiated to develop numerical reactors (Chauliac et al., 2011; Stanek, 2019; Szilard et al., 2011). This technology leverages high-fidelity numerical simulation methods, which are renowned for their precision, resolution, and reliability, to reduce unnecessary conservatism. This approach facilitates accurate prediction of essential thermal–hydraulic parameters in reactor systems across a wide range of operating conditions (Wu et al., 2022). By leveraging data-driven methods, the development of thermal–hydraulic models lays a solid foundation for addressing flow and heat transfer challenges in numerical reactors. By harnessing the vast computational data generated by numerical reactors, these data-driven thermal–hydraulic models can undergo continuous updates and iterative optimizations, ultimately enhancing their versatility and computational precision.

This method, based on deep learning algorithms, seeks to establish relationships among internal features within massive datasets. Among various data-driven methods, Artificial Neural Networks (ANN) have become one of the most widely applied methods due to their excellent ability to solve nonlinear problems. Introduced by Warren McCulloch and Walter Pitts in 1943 (Mcculloch and Pitts, 1943), ANN is a complex network structure formed by numerous interconnected neurons that transmit information through activation functions and weight coefficients. ANN possesses advantages such as self-learning, adaptability, and robustness, enabling rapid and accurate solutions to complex nonlinear problems.

Given that ANN can summarize regularities and predict results in situations where internal mechanisms and physical relationships are not entirely clear, many scholars globally have applied ANN to predict Critical Heat Flux (CHF). Moon and Chang (Moon and Chang, 1994) first classified CHF data using fuzzy theory and then analyzed it using neural networks. Lee et al. (Lee et al., 2000) applied ANN and regression algorithms to correct the KAIST critical heat flux database, investigating the influence of aspect ratio on critical heat flux. Vazri et al. (Vaziri et al., 2007) separately established neural network models based on radial basis functions and multilayer perceptron, training them with CHF experimental data. They found that radial basis neural networks exhibited superior predictive capabilities for CHF. Nafey (Nafey, 2009) developed a CHF prediction model based on ANN, using the Levenberg-Marquardt algorithm to train the developed feedforward neural network. This method outperformed traditional lookup tables and empirical correlations, providing a new CHF calculation formula. Rostamifard et al. (Rostamifard et al., 2011) developed CHF prediction models based on mixed-order neural networks and feedforward neural networks. Comparative analysis revealed that mixed-order neural networks achieved higher accuracy in CHF calculations. Ertunc et al. (Ertunc, 2006) established a pool boiling CHF prediction model based on ANN, demonstrating high predictive accuracy for pool boiling CHF. Han et al. (Han, 2005) predicted CHF using ANN theory based on data from uniformly heated vertically ascending circular tubes. Su et al. (Su et al., 2002) used three different neural network models to predict CHF. Wei et al. (Wei et al., 2009, 2010; Wei et al., 2009) applied wavelet neural networks to analyze CHF, two-phase flow instability, and transition boiling. Zhao et al. (Zhao et al., 2020) applied physics-informed machine learning-aided framework to analyze CHF, achieving positive outcomes in these studies.

Current research mostly focuses on the development and training of models using experimental data, and their application is typically limited to comparisons with experimental data. The integration of models with thermal–hydraulic codes has not been widely explored. To establish a high-precision, wide-parameter-range, and highly adaptable CHF prediction model and integrate it into a system analysis code for accurate CHF predictions during transient reactor conditions, this study utilized a CHF lookup table (Groeneveld et al., 2007) and experimental data (Kim et al., 2000) to establish a comprehensive CHF database. The training data were normalized, and CHF prediction models were separately developed based on the Backpropagation Neural Network (BPNN), Random Forest (RF), and Physics-informed Machine Learning (PIML) models. After training, the models were compared based on prediction errors and computational time to select the optimal CHF prediction model. The best CHF prediction model was then integrated into self-development analysis code ARSAC (Deng et al., 2021), replacing the original CHF calculation model. This model can continuously improve prediction accuracy and expand its applicability range through supplementary training data. Finally, the accuracy and feasibility of the modified code were verified through international standard benchmark tests such as the Oak Ridge National Laboratory (ORNL) rod bundle heat transfer experiment, achieving the realization of data-driven continuous improvement in CHF prediction technology. This method can provide support for the prediction of key thermal–hydraulic parameters in nuclear reactors based on numerical reactor technology.

Table 1
Common CHF correlations and applicable ranges.

Correlation	P/MPa	G/ (kg·m^{-2}·s^{-1})	D_H/ mm	x	L_H/m
B&W-2 (Agee, 1997)	13.79 ~ 16.55	1020 ~ 5425	5.1 ~ 12.7	−0.03 ~ 0.2	1.828
W-3 (Tong, 1967)	6.895 ~ 16.55	1356 ~ 6780	5.08 ~ 17.8	−0.15 ~ 0.15	0.25 ~ 3.67
Barnett (Barnett, 1966)	1.034 ~ 5.0	40.69 ~ 2305.59	10 ~ 13.8	——	0.836 ~ 4.44
Biasi (Biasi, 1967)	0.27 ~ 14.0	100 ~ 6000	3 ~ 37.5	0 ~ 1	0.2 ~ 6
WRB-1 (Motley et al., 1976)	9.9 ~ 17.2	777.8 ~ 3250	9.4 ~ 15.2	−0.2 ~ 0.3	0 ~ 4.27
Weisman (Weisman and Pei, 1983)	2.0 ~ 20.5	972.2 ~ 13611	1.15 ~ 37.5	0 ~ 0.6	0.35 ~ 3.6

2. Mathematical models

2.1. Backpropagation neural network (BPNN) model

Artificial Neural Network (ANN) is a computational structure based on simulating the mechanism of biological neurons, representing an abstraction and simplification of biological neurons. Generally, a neural network comprises an input layer, one or multiple hidden layers, and an output layer. Each layer contains information processing units capable of implementing fundamental functions akin to those of biological neurons. These information processing units in a neural network are referred to as neurons or nodes (Zhang et al., 2019).

The mathematical model is as follows:

$$o_j(t) = f\left(\sum_{i=1}^{n} \omega_{ij} x_i(t - \tau_{ij}) - T_j\right) \quad (1)$$

Where: $o_j(t)$ represents the information output of neuron j at time t, τ_{ij} is the synaptic delay between input and output, T_j is the threshold of neuron j, ω_{ij} is the weight factor from neuron i to j, $f(x)$ is the activation function of the neuron.

Artificial neural networks can be categorized into various types depending on the connection patterns and weight optimization methods. These types include Backpropagation Neural Network (BPNN), Wavelet Neural Network (WNN), Genetic Neural Network (GNN), and Recurrent Neural Network (RNN).

In this study, a four-layer BPNN was chosen to train the CHF model. The typical structure of a four-layer BPNN is illustrated in Fig. 1, consisting of an input layer, two hidden layers, and an output layer.

The BPNN is a multilayer feedforward network trained through the process of error backpropagation, with its fundamental principle rooted in the gradient descent method. The core idea involves utilizing gradient search techniques to minimize the mean squared error between the actual network output and the expected output. The basic BP algorithm consists of two primary processes: forward signal propagation and backward error propagation.

During the forward propagation phase, computations proceed from input to output, with input signals traversing through hidden layers to influence the output nodes. Through nonlinear transformations, these signals generate output signals. In cases where the actual output deviates from the expected output, the process transitions to the error backpropagation phase. In error backpropagation, output errors systematically propagate backward through the hidden layers, starting from the output layer and progressing toward the input layer. The adjustment of weights and thresholds occurs in the reverse direction, from the output layer to the input layer. The iterative nature of this process aims to minimize errors at the output layer by distributing errors among all units in each layer. The error signals obtained from each layer serve as the basis for adjusting the weights of individual units.

In summary, the BPNN employs the gradient descent method to iteratively adjust weights and thresholds, minimizing the mean squared error between actual and expected outputs. The forward propagation phase allows input signals to traverse through the network, and in cases of disparity, the error backpropagation process systematically adjusts network parameters to enhance overall performance.

2.2. Random forest model

The Random Forest (RF) model was introduced by Breiman in 2001 (Breiman, 2001). A Random Forest entails constructing numerous decision trees, each employing distinct partitioning methods. The structure of a Random Forest is depicted in Fig. 2.

The process entails training each decision tree separately on randomly sampled subsets of the entire training data. Subsequently, the results obtained from each decision tree are compared and evaluated. During prediction, the data to be predicted is subjected to each decision tree individually, and the final result is determined through a voting process based on the predictions of each decision tree.

Random Forest effectively addresses the issue of overfitting. The parameters related to the decision tree component in Random Forest are similar to those of a standalone decision tree model. Additionally, Random Forest model parameters include the number of decision trees,

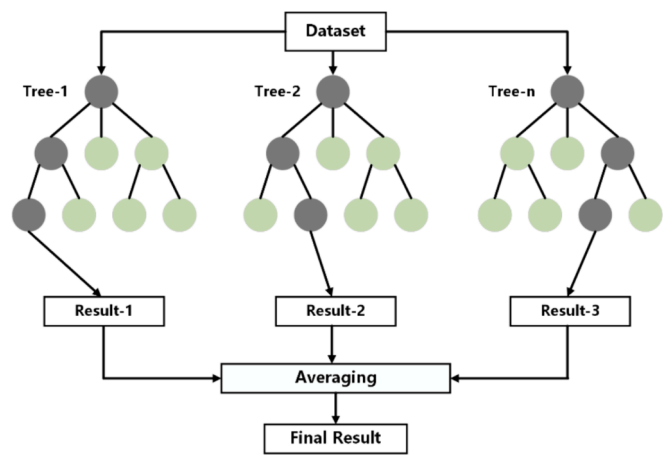

Fig. 2. The structure of RF.

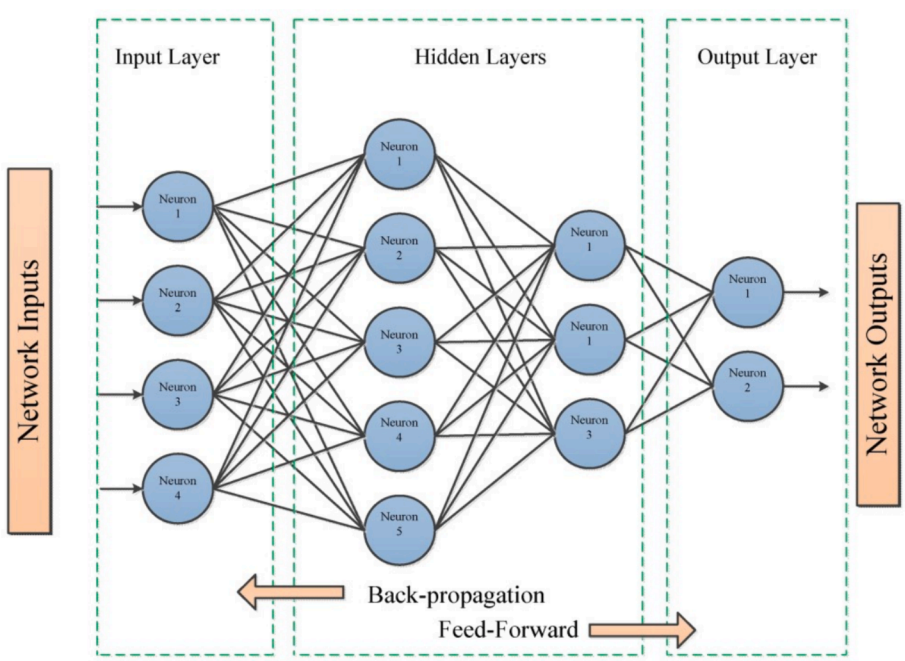

Fig. 1. The structure of BPNN.

whether sampling is conducted with or without replacement, whether out-of-bag samples are used for model evaluation, and the maximum number of features. Among these parameters, the key ones are the number of decision trees and the maximum number of features. The latter is typically set to the logarithm base 2 of the sample size or the square root of the sample size.

2.3. Physics-informed Machine Learning (PIML) model

PIML is a neural network algorithm with physical information constraints, which is usually suitable for solving physical problems where differential equations cannot be established. The structure of PIML is illustrated in Fig. 3.

PIML initially computes the input data utilizing a prior model to obtain the difference between the prior model and the actual value. Subsequently, the neural network undergoes training, with the difference as the training objective to obtain the relationship between the input data and the difference. The output of the final model is determined by the output of the neural network and the output of the prior model. Within this algorithm framework, the principal role of the neural network model is to correct the physical relationship in the prior model, so the final output result of the model is more consistent with the description of the physical relationship.

The advantage of this model is that it can be learned from the residual between the physical relationship and the actual result. Moreover, the applicability of the physical relationship is not limited solely to the physical field problem that needs to be defined by the differential equation, and it extends to encompass a broader array of empirical relationships commonly encountered across various disciplines.

2.4. Data processing

The collected data is sourced from the CHF lookup table and literature data, comprising a total of 7,758 data sets. The parameter range of the data is presented in Table 2.

As the CHF lookup table is established based on an 8 mm circular tube, the actual CHF values need to be corrected according to the specific hydraulic diameter of the flow channel. The correction formula is as follows (Groeneveld et al., 2007):

$$\begin{aligned}&\mathrm{CHF}\left(D_{\exp}, p_{\exp}, G_{\exp}, x_{\exp}\right)\\&=\mathrm{CHF}\left(D=8, p_{\exp}, G_{\exp}, x_{\exp}\right)\left(\frac{D_{\exp}}{8}\right)^{-1/2}\end{aligned} \tag{2}$$

Where $D_{\exp}$ is the hydraulic diameter of the actual flow channel (mm), $p_{\exp}$ is the actual pressure (kPa), $G_{\exp}$ is the actual mass flux ($kg{\cdot}m^{-2}{\cdot}s^{-1}$), and $x_{\exp}$ is the actual quality.

Table 2
Parameter ranges for the CHF data.

Author	P/MPa	G/ ($kg{\cdot}m^{-2}{\cdot}s^{-1}$)	D_H/ mm	x	Data size
Groeneveld (Groeneveld et al., 2007)	0.1 ~ 21	0 ~ 8000	8	−0.5 ~ 1.0	7245
Kim (Kim et al., 2000)	0.106 ~ 0.951	20 ~ 277	6 ~ 12	0.653 ~ 0.732	513

Due to the significant differences in the ranges of various parameter values, in order to ensure that all features have a consistent measurement scale, normalization is performed on all input and output data before training. The normalization method is as shown in Equation (3):

$$\mathrm{X}_{\mathrm{norm}}(i)=\frac{\mathrm{X}(i)-\mathrm{X}_{\min}}{\mathrm{X}_{\max}-\mathrm{X}_{\min}} \tag{3}$$

Where $\mathrm{X}_{\mathrm{norm}}$ represents the normalized data, X is the original data, $\mathrm{X}_{\min}$ is the minimum value in the same category of data, and $\mathrm{X}_{\max}$ is the maximum value in the same category of data.

During the process of model training, the variables encompassing pressure, mass flow rate, hydraulic diameter, and void fraction are utilized as input parameters, thereby forming the set of training inputs. Simultaneously, the output variable considered for training is the critical heat flux.

3. Training and evaluation of data-driven models

3.1. The model based on BPNN

The development and training of the BPNN model are conducted using Python coding, leveraging the Keras library for constructing the neural network architecture (Gulli and Pal, 2017). A four-layer neural network, including one input layer, one output layer, and two hidden layers, was established, with the numbers of neurons in the two hidden layers being the same. To mitigate the risk of overfitting, the Dropout technique was introduced (Hinton et al., 2012). The input layer comprises three neurons representing pressure, mass flow density, and quality, while the output layer consists of a single neuron responsible for predicting the Critical Heat Flux (CHF), utilizing the ReLU (Rectified Linear Unit) activation function.

Given the substantial impact of the number of neurons in the hidden layers on network quality, the hidden layer neuron count is varied from 20 to 100 during neural network training. To assess network quality, the Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) are chosen as metrics. The calculation formulas for these metrics are expressed in Equations (4) and (5), respectively:

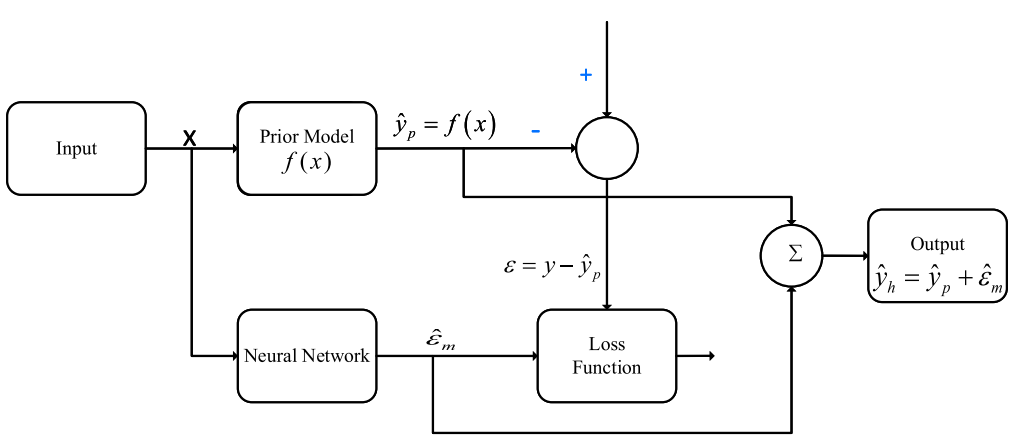

Fig. 3. The structure of PIML.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\mathrm{CHF}_{act}(i) - \mathrm{CHF}_{pre}(i)}{\mathrm{CHF}_{act}(i)}\right| \times 100\% \tag{4}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}\left(\mathrm{CHF}_{act}(i) - \mathrm{CHF}_{pre}(\mathrm{i})\right)^2}{n}} \tag{5}$$

The collected data is divided into training and testing sets in a ratio of 8:2, utilizing 6206 data points for training and 1552 data points for testing the pre-trained neural network model. To choose the optimal number of neurons in the hidden layers, k-fold cross-validation was adopted (Burman, 1989). The procedure is as follows:

a) Shuffle the dataset randomly;
b) Split the dataset into k groups (k = 5);
c) For each neuron count, repeat the training five times, choosing one group as the testing set and the other four groups as the training set each time;
d) Compute the average MAPE and RMSE of the testing set for each neuron count.

The influence of different hidden layer neuron counts on the error is summarized, as depicted in Fig. 4. From Fig. 4, it is observed that the neural network's error decreases with an increasing number of hidden layer neurons and gradually stabilizes. Beyond a hidden layer neuron count of 60, there is no significant reduction in error. Considering that an excessive number of hidden layer neurons can complicate the neural network structure, increasing the model's training and computation time, the hidden layer neuron count is set to 60.

After 2647 training iterations, the MAPE on the test set reached 18.01 %, which is consistent with the expected results, signifying the completion of the model training. Subsequently, the fully trained neural network model was employed to predict all the collected data, with a total prediction time of 4.4 s. Error calculations were performed, and Fig. 5 illustrates the dispersion of errors in the predictions generated by the BPNN model. It can be observed that the prediction errors predominantly fall below 20 %, and as the CHF increases, exhibiting a gradual decline in relative error with increasing CHF.

3.2. The model based on RF

When developing the CHF prediction model using the Random Forest (RF) model, a random forest model with 200 decision trees was established. The parameters for each decision tree in the forest were set with the maximum number of features for selection equal to the number of samples, the minimum number of samples in a leaf node as 1, and in an internal node as 2. No constraints were imposed on the maximum number of leaf nodes or the maximum depth, ensuring comprehensive partitioning of each decision tree's nodes.

Simultaneously, the collected dataset was partitioned into training and testing sets in 8:2 ratio. The evaluation metric for the prediction results of the constructed decision trees in the forest was set to the Absolute Error (AE), defined according to Equation (6):

$$AE = \left|X_{predict} - X_{true}\right| \tag{6}$$

Following the training phase, the developed model demonstrated an average relative error of 5.23 % in predicting the test set, indicating the completion of the training process. Subsequently, the model was employed to predict the entire training set, with a total computation time of 257 s. The calculated relative errors are depicted in Fig. 6, illustrating that the prediction errors are predominantly below 10 %. Furthermore, a discernible trend is observed where the relative error diminishes gradually with an increase in CHF.

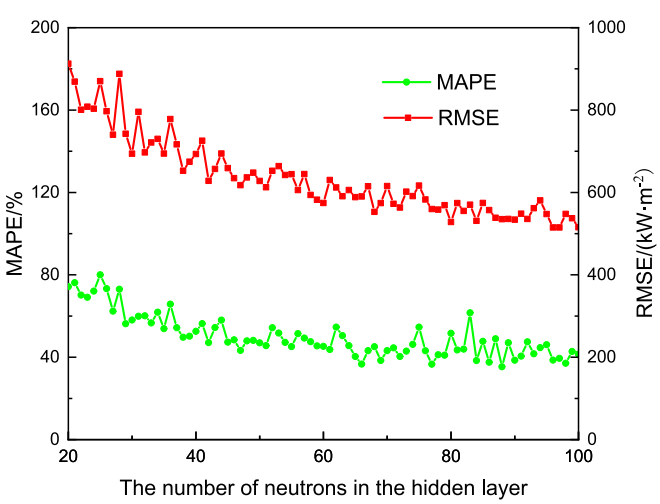

Fig. 4. Influence of hidden layer neuron number on prediction errors of neural network.

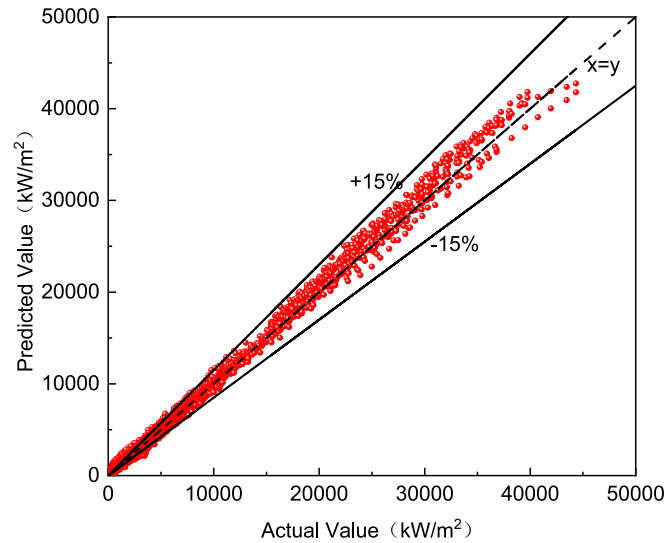

Fig. 5. Calculation error distribution of BPNN.

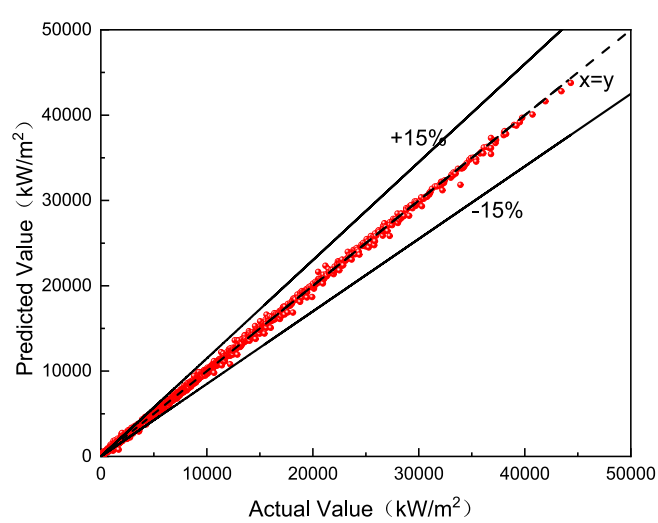

Fig. 6. Calculation error distribution of RF.

3.3. The model based on PIML

When the CHF prediction model was developed based on PIML, the prior model is Biasi correlation and the neural network model is MLP. Since the scope of application of Biasi correlation is smaller than the collected data set, the data set is filtered and the data within the scope of application of the correlation is retained as training and validation data. The number of hidden layer neurons selected in MLP is 100 and the number of hidden layer is 2.

The collected data were divided into training set and test set according to 8:2. What's more, MAPE and RMSE were still selected as metrics to measure the quality of the model. After training, the average relative error of the test set reached 18.34 %, which was consistent with the expected result, and the model training was completed. At this time, the trained PIML model was used to re-predict all the collected data. The total prediction time was 8.23 s, and the error calculation was carried out. The MAPE was 18.80 %. The prediction results of PIML are depicted in Fig. 7, the prediction error distribution of the model is relatively uniform.

3.4. Comparison of the models

The comparison of MAPE and computation times between the three models is illustrated in Fig. 8. It is evident that the BPNN exhibits faster computational speed, while the random forest model achieves higher computational accuracy. The extended computation time of the random forest model is attributed to its inherently complex structure. The calculation accuracy of PIML model is close to that of BPNN, but the calculation time is 1.8 times that of BPNN. Conversely, the error in the BPNN can be further reduced with the addition of subsequent training data. Therefore, the BPNN model is selected as the optimal model for CHF prediction due to its favorable balance of accuracy and computational efficiency.

3.5. Comparison with traditional correlations

In the analysis of the developed Critical Heat Flux (CHF) model, data points within the applicable ranges of the W-3 and Biasi correlations were selected respectively. Comparative calculations were performed using these correlations and the data-driven model.

Specifically, 309 data points falling within the applicable range of the W-3 correlation were utilized. Calculations were performed using both the W-3 correlation and the neural network model.

Figs. 9 and 10 present the calculation results derived from the W-3 correlation and the data-driven model, respectively, for the selected data points. It is indicated that the results from the data-driven model are more consistently within the ± 20 % error margin, demonstrating that the data-driven model offers superior precision compared to traditional empirical correlations.

The W-3 correlation produced an average relative error of 12.92 % when applied to the dataset. In contrast, the data-driven model achieved a significantly lower average relative error of 6.12 %. This represents a precision improvement of 6.8 %, underscoring the enhanced accuracy of the data-driven model over the W-3 correlation.

Within the applicable range of the Biasi correlation, 264 data points were selected for comparative analysis with the computation results of the data-driven model. Fig. 11 illustrates the computation results using the Biasi correlation, while Fig. 12 presents the corresponding results from the data-driven model.

According to the results, it is observable that the computation results generated by the Biasi correlation exhibit error margins extending to ± 40 %, while the results generated by the data-driven model are primarily concentrated within a ± 20 % range. This suggests a significant improvement achieved by the data-driven model over the Biasi correlation. Specifically, the MAPE of the dataset computed using the Biasi

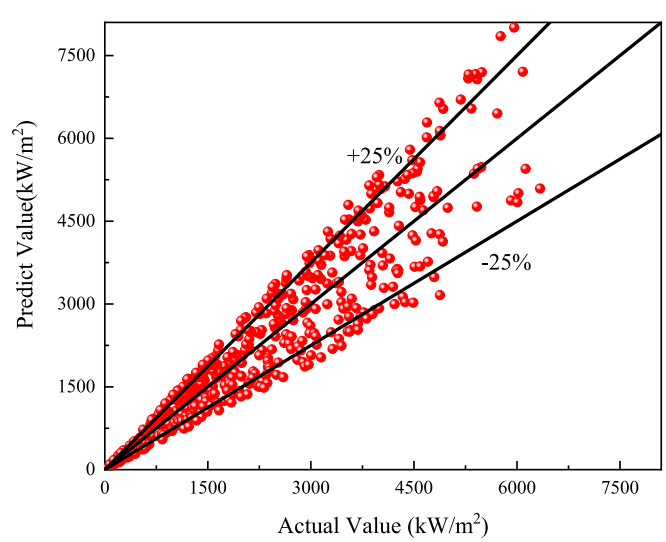

Fig. 7. The prediction results of PIML.

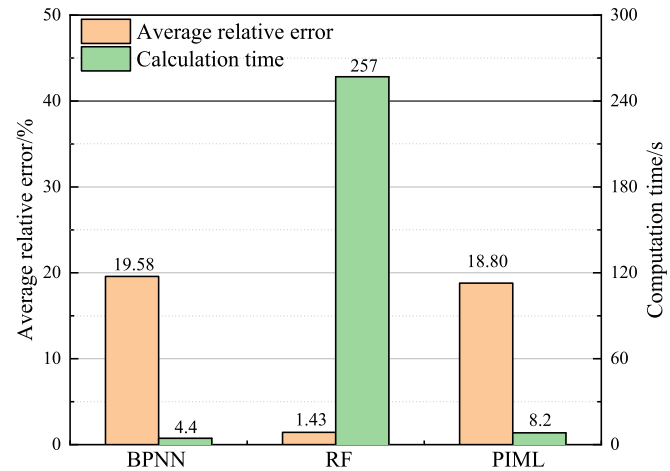

Fig. 8. Comparison of the three models.

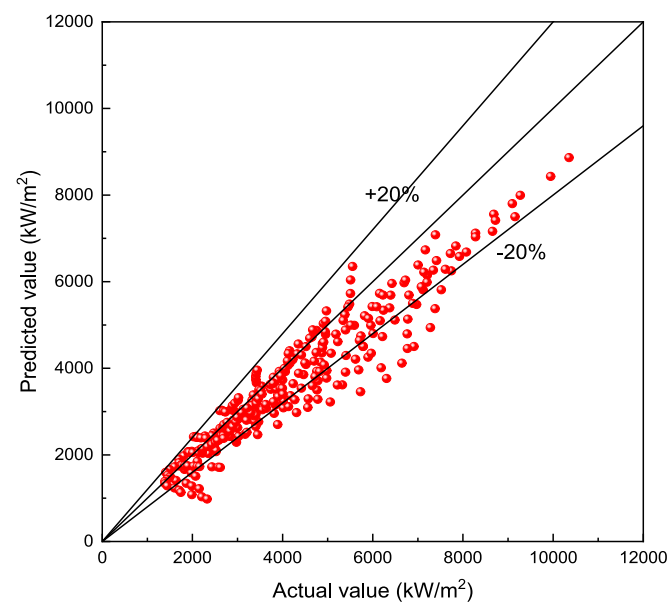

Fig. 9. The prediction results of W-3 correlation.

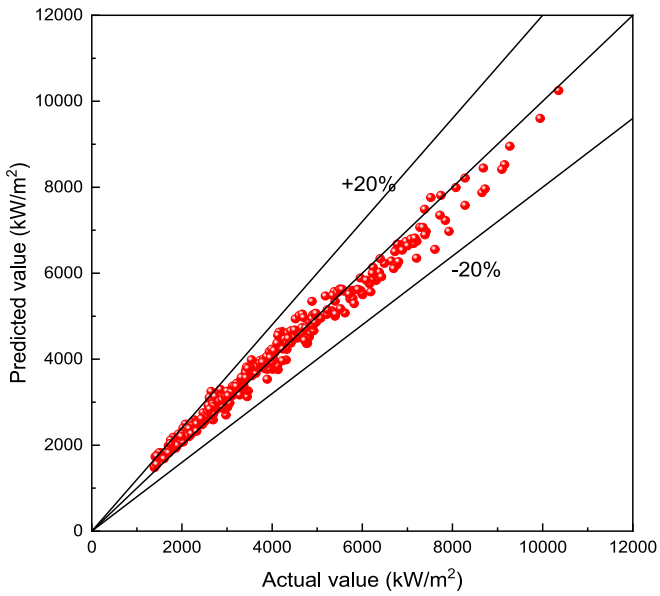

Fig. 10. The prediction results of data-driven model.

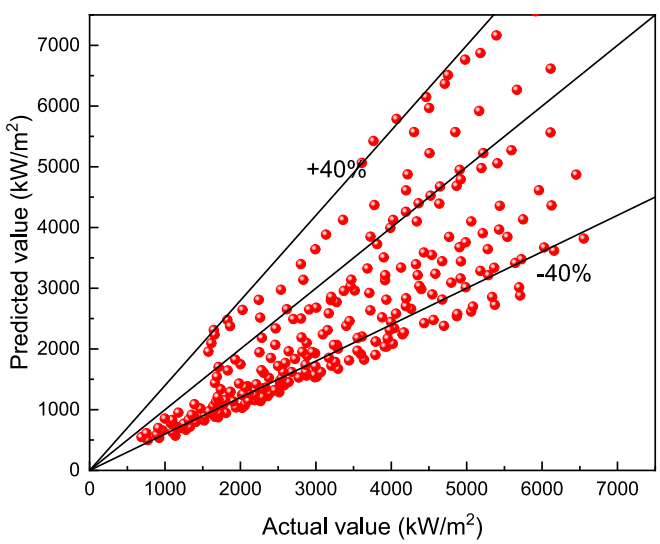

Fig. 11. The prediction results of Biasi correlation.

correlation is 30.8 %, while the data-driven model achieves a markedly lower MAPE of 6.3 %, representing a substantial error reduction of 24.5 %.

4. Validation of data-driven CHF model in system analysis code

4.1. Implement of system analysis code

The external coupling approach between the Fortran-based system analysis code and the Python-based data-driven model is time-consuming due to data exchange. To enhance computational efficiency while maintaining calculation accuracy, an internal coupling approach based on the data-driven Critical Heat Flux (CHF) model was applied to modify the system analysis code. The specific coupling steps are shown in Fig. 13.

1) Integration of the data-driven CHF model algorithm module into the system analysis code: The algorithm module based on the data-driven CHF model was implemented in the system analysis code, and a complete computation subroutine based on the BPNN was established in this module.

2) Acquisition of structural parameters for the data-driven CHF model: The various structural parameters of the data-driven CHF model were obtained, including the number of layers in the neural network, the number of neurons in each layer, weights, and biases. With all these structural parameters and inputs, CHF can be calculated.

3) Transfer of model structural parameters during input card reading: While reading the input file, the structural parameters of the neural network model were read and stored in common variables of the system analysis code.

4) Reconstruct the CHF model in the system analysis code: A subroutine for CHF calculation was built, which is called directly for CHF calculation as needed. The structural parameters stored in the common variables are used during the CHF calculation based on the neural network.

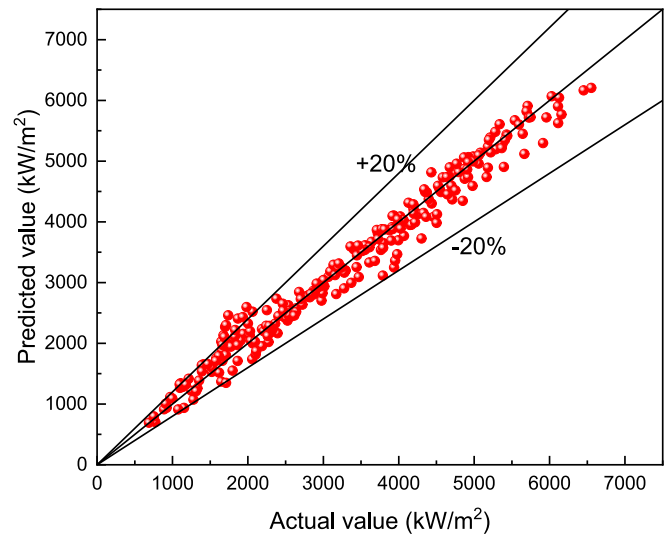

Fig. 12. The prediction results of data-driven model.

4.2. Introduction of ORNL experiment

This study validated the code using rod bundle heat transfer experimental data from the Oak Ridge National Laboratory (ORNL) Thermal-Hydraulic Test Facility (THTF) (Yoder et al., 1985). The experimental loop and the cross-section of the test section are illustrated in Fig. 14 and Fig. 15, respectively.

The coolant, pressurized by a pump, flows through a horizontal pipe and a downcomer before entering the lower plenum. Ascending from the bottom, it flows through the heated test section and exits via the upper plenum. In the heat exchanger section, the coolant is cooled to the desired inlet temperature for the test section, thereby establishing a closed loop. A pressurizer is incorporated into the experimental loop to maintain control over the system pressure. The test section comprises 64 rods arranged in an 8 × 8 array, with 60 rods serving as heating rods and 4 rods as non-heating rods simulating control rods. The specific rod bundle structural parameters are summarized in Table 3. Additionally, valves simulating Loss of Coolant Accident (LOCA) events are placed at the inlet and outlet positions of the test section in the experimental loop.

The ORNL-THTF rod bundle heat transfer experiment consists of 30 steady-state conditions, numbered from B to X and IA to IH, and 3 transient conditions, numbered from 3.06.6A to 3.06.6C. In this study, the ID condition from the steady-state conditions is selected for comparison. The initial operation parameters for the ID condition are as follows: heating rod heat flux density of 275 kW/m^2, mass flow density of 46 kg/(m^2·s), inlet pressure of 8.06 MPa, and test section inlet temperature of 562 K.

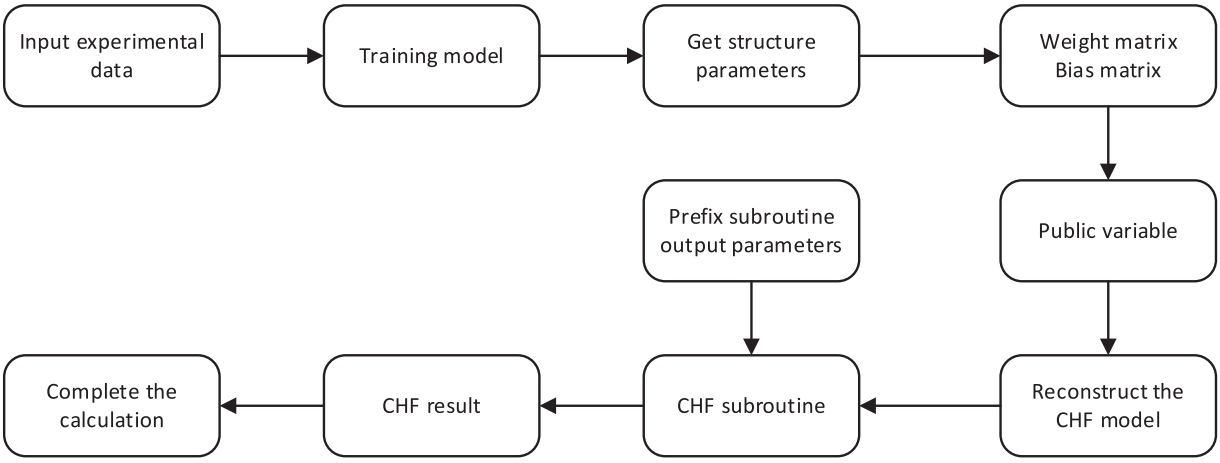

Fig. 13. CHF model coupling flow chart.

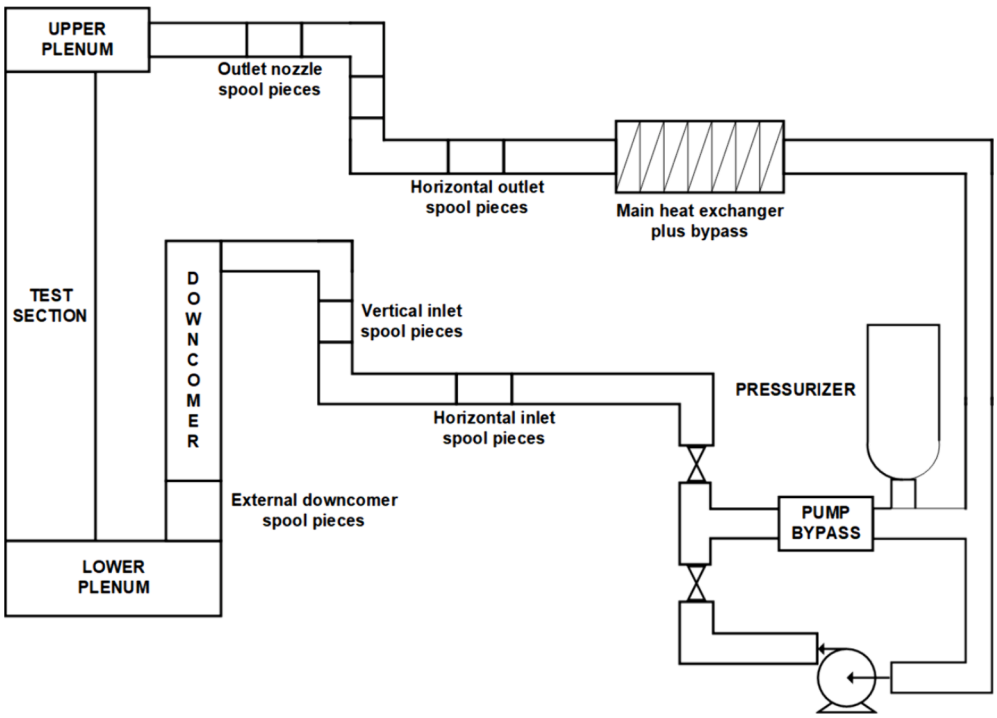

Fig. 14. ORNL-THTF experimental device diagram (Yoder et al., 1985).

4.3. Validation results

To model the experimental loop, a nodal diagram is presented in Fig. 16. The component labeled 110P represents the test section, which is divided into 20 control volumes, each with a length of 0.183 m. The total length of component 110P is 3.66 m, and the flow area is 0.00624 m^2. Component 108P serves as the heat exchanger, maintaining the inlet temperature of the test section at 562 K. Component 126P is a simulated pressurizer controlling the inlet pressure of the test section, while 113PMP represents a simulated pump controlling the fluid flow in the loop. Through calculations, the mass flow rate in the loop is set to 0.91104 kg/s, and the total heating power in the test section is set to 1.8 MW.

Calculation of the experimental loop was performed using both the original and implemented code, and the results were compared with the experimental wall temperature data from the literature, as shown in Fig. 17.

According to the calculation results, there is no significant increase in the pipe wall temperature before reaching the CHF. The results obtained from the original code, the implemented code, and the experiment are consistent. After reaching the CHF, the wall temperature rises sharply, transitioning from nucleate boiling to film boiling.

Compared to the original code, the implemented code decreased the average calculation error from 22.84 % to 3.64 %. This average calculation error represents the average relative error between experimental data points and calculation results at corresponding positions. Compared with the actual experimental data, the results calculated by the original code were higher, while the results from the implemented code matched well. This indicates that the code based on a data-driven model achieved higher accuracy in wall temperature calculation. However, during the process of wall temperature sharp rise, the wall temperature calculated by implemented code were lower. This is mainly due to the fact that the implemented code calculated the onset of heat transfer deterioration higher than the actual, resulting in a delayed

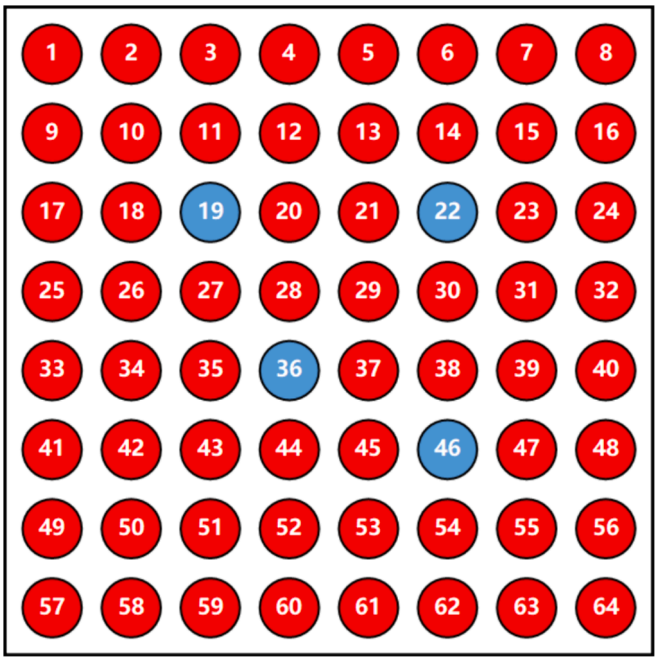

Fig. 15. The section diagram of ORNL-THTF test section (Yoder et al., 1985).

Table 3
Parameters of rod bundle test section.

Parameter	Value
The number of heated rods	60
The number of unheated rods	4
The diameter of heated rods	0.95 cm
The diameter of unheated rods	1.02 cm
Pitch	1.27 cm
The cross-area of test section	0.104 m × 0.104 *m*
The length of test section	3.66 *m*
Hydraulic diameter	1.06 cm
Heating equivalent diameter	1.38 cm

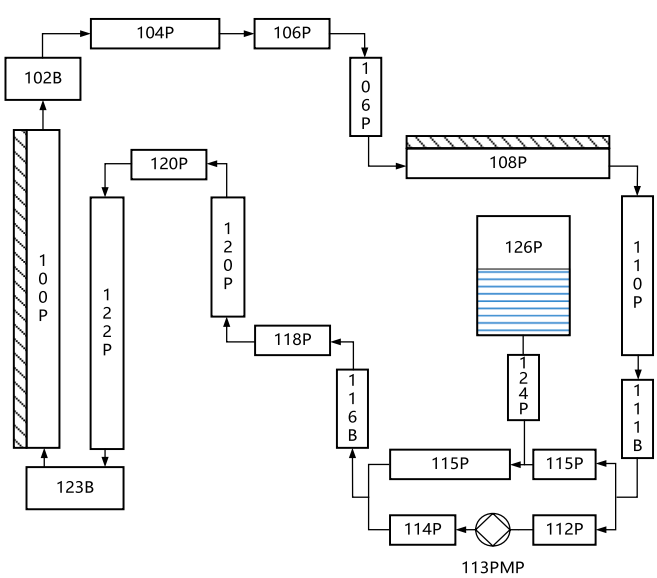

Fig. 16. Model of the experiment loop.

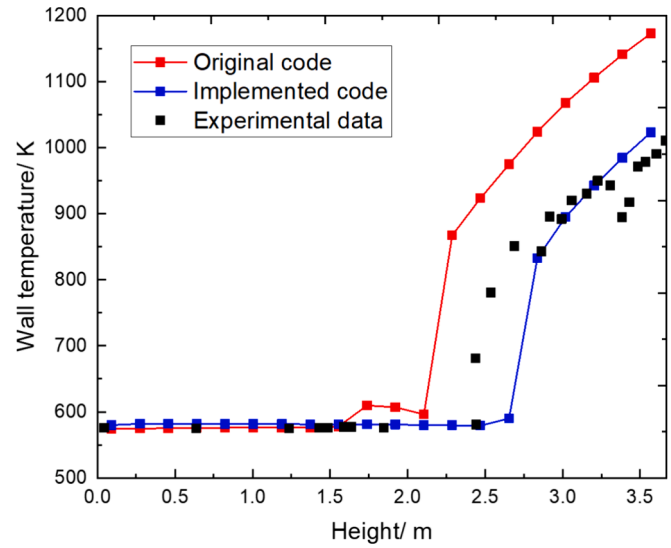

Fig. 17. Comparison of calculated wall temperature with experimental data.

occurrence of wall temperature sharp rise.

As the literature only provides wall temperature data, the results of other parameters from the implemented code were compared with the original code. The comparative results are shown in Figs. 18 to 21.

Fig. 18 illustrates the comparison of CHF at different axial heights. As the coolant in the pipe is continuously heated, bubbles gradually form in the single-phase liquid at the inlet. Due to the intense disturbance caused by the generation and detachment of bubbles on the wall, CHF gradually decreases with increasing pipe height until it reaches 0 after complete transition to film boiling. The results indicate that the CHF values from the implemented code are higher, suggesting that the implemented code has further compressed the safety margin provided by the original code.

Figs. 19-21 show the comparison results of void fraction, liquid phase velocity, and gas phase velocity at different axial heights, respectively. From the comparison results, it can be observed that the differences in the calculation of these parameters between the original and implemented codes are small, indicating that the modification to the CHF model did not have a significant impact on the calculation of other important parameters. As shown in the results, the impact starts to become apparent at a void fraction of about 0.8. This is because a two-phase flow model is adopted in the code, and the heat transfer coefficients to liquid and vapor are calculated separately. Additionally, the

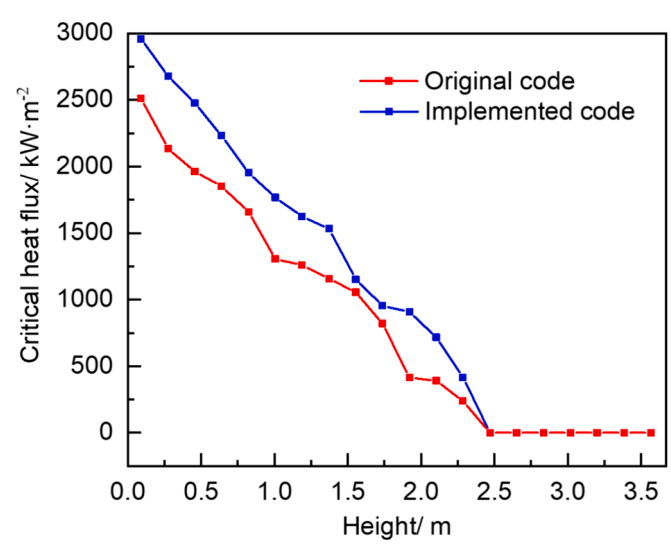

Fig. 18. Comparison results of CHF.

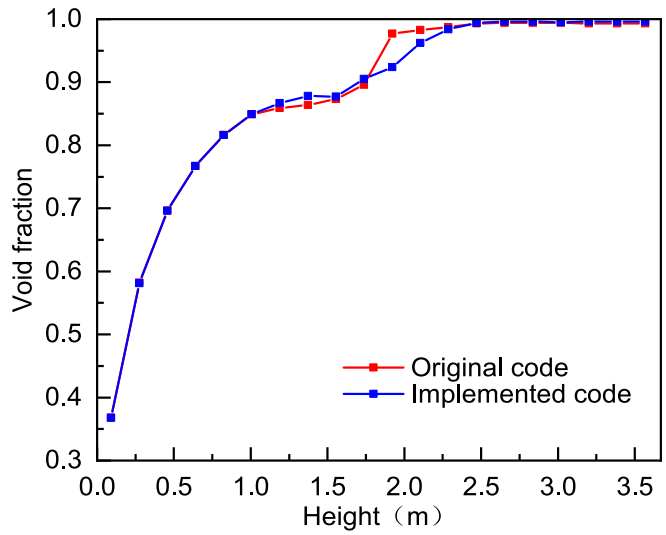

Fig. 19. Comparison results of void fraction.

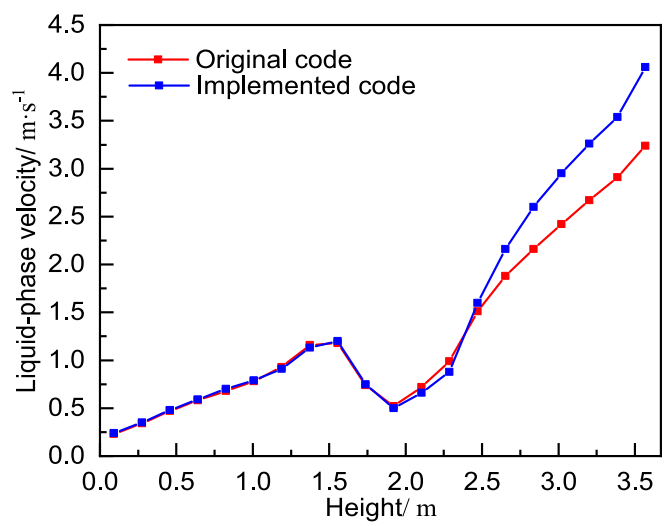

Fig. 20. Comparison results of liquid velocity.

heat transfer coefficient to vapor is related to the total wall heat flux and critical heat flux at high void fractions. The implemented code still exhibits good calculation stability.

5. Conclusion

This paper presents the development of a data-driven Critical Heat Flux (CHF) model and its integration into ARSAC. The key findings are outlined below:

(1) A comprehensive CHF database was established, incorporating CHF lookup tables and experimental data, with normalization of the training data. BPNN, RF and PIML models were utilized to establish CHF prediction models. A comparative analysis led to the selection of the BPNN as the superior CHF prediction model, achieving an overall prediction deviation of under 20 %.

(2) The optimized CHF prediction model was successfully integrated into the system analysis code, superseding the existing CHF calculation model. This updated model boasts the ability to enhance prediction accuracy and broaden its applicability through additional training data.

(3) The modified code was validated using the ORNL rod bundle heat transfer experiment. Compared to the original code, the updated version reduced the deviation in predicting wall temperatures from 22.84 % to 3.64 %, significantly highlighting its precision and practicality. This approach paves the way for continuous advancement in CHF prediction technology based on data-driven methods.

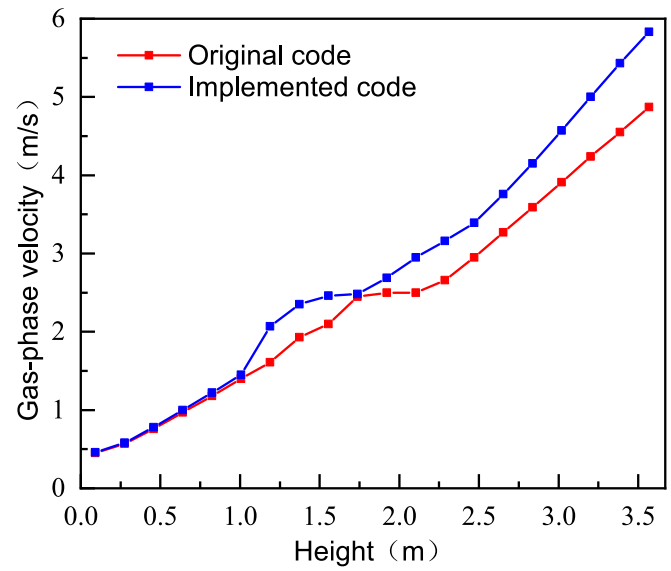

Fig. 21. Comparison results of gas velocity.

This methodology offers a valuable tool for predicting critical thermal–hydraulic parameters of nuclear reactors through numerical reactor simulations. Future advancements in prediction accuracy are anticipated through the expansion of CHF training data and the refinement of neural network architectures.

CRediT authorship contribution statement

Zhifang Qiu: Conceptualization. **Yichao Ma:** Writing – original draft. **Tao Huang:** Writing – review & editing. **Jian Deng:** Methodology. **Dexiang Kong:** Methodology. **Dan Wu:** Validation. **Jing Zhang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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